

ORIGINAL ARTICLE

Special Section: Tribute to Rien van Genuchten, Recipient of the 2023 Wolf Prize for Agriculture

Coupled hydro-mechanical pore-scale modeling of biopore-coated clods for upscaling soil shrinkage and hydraulic properties

Luis Alfredo Pires Barbosa  | Horst H. Gerke

Research Area 1 “Landscape Functioning”,
Working Group “Silicon Biogeochemistry”,
Leibniz Centre for Agricultural Landscape
Research (ZALF), Müncheberg, Germany

Correspondence

Luis Alfredo Pires Barbosa, Research Area
1 “Landscape Functioning”, Working Group
“Silicon Biogeochemistry”, Leibniz Centre
for Agricultural Landscape Research
(ZALF), Eberswalder Strasse 84, 15374,
Müncheberg, Germany.
Email: Luis.Pires@zalf.de

Assigned to Associate Editor Chao Zhang.

Funding information

Deutsche Forschungsgemeinschaft,
Grant/Award Number: GE 990/14-1

Abstract

Earthworms and plant roots are vital for macropore formation and stabilization. The organo-mineral coating of biopore surfaces also regulates macropore-matrix mass exchange during preferential flow. The influence of finer-textured burrow coatings on macroscopic soil properties during shrinkage could potentially be assessed by upscaling pore-scale hydraulic and mechanical simulations. The aim was to investigate the influence of micro parameters (particle size, stiffness, and bond strength) on macro parameters (i.e., shrinkage curve and soil hydraulic properties). Drainage experiments and simulations were carried out using biopore-coated clod-size samples compared to those without coating. Simulations were performed using a two-phase pore-scale finite volume coupled with discrete element model (DEM-2PFV). The structural dynamics was characterized by analyzing the pore volume and soil shrinkage curve obtained from numerically determined data. The soil hydraulic parameters were described using uni- and bimodal van Genuchten (vG) functions. The drainage simulations revealed hydro-mechanical dynamics characterized by Braudeau-shrinkage curve subdomains: The matrix-only samples, with lower particle bond strength, exhibited relatively higher shrinkage. The coated samples, with higher particle stiffness and bond strength, displayed greater hydro-mechanical stability. The numerically determined initial value of the saturated hydraulic conductivity (K_s) was about 70 times larger for matrix-only samples than for coated samples. As expected, for the nonrigid soil structures, constant K_s , α , and n values for bimodal vG model resulted in prediction errors. Upscaling DEM-2PFV pore-scale model outcomes quantifies micro-coating effects on macro hydro-mechanics. This yields void ratio-based soil water retention and hydraulic conductivity functions, advancing macroscopic soil hydraulic models and enhancing structured soil flow and transport descriptions.

Abbreviations: DEM, discrete element model; K_s , hydraulic conductivity; vG, van Genuchten.

This is an open access article under the terms of the [Creative Commons Attribution-NonCommercial-NoDerivs](https://creativecommons.org/licenses/by-nc-nd/4.0/) License, which permits use and distribution in any medium, provided the original work is properly cited, the use is non-commercial and no modifications or adaptations are made.

© 2024 The Authors. *Vadose Zone Journal* published by Wiley Periodicals LLC on behalf of Soil Science Society of America.

1 | INTRODUCTION

At field and pedon scale (Figure 1), the soil hydraulic properties characterizing variably saturated flow have often been described using unimodal (van Genuchten, 1980) or bimodal functions (Durner, 1994), among other more advanced functions (Seki et al., 2022). The choice of these macroscopic soil hydraulic functions depends on various factors, including the scale of observation (Vereecken et al., 2007). However, current models that characterize soil hydraulic properties at field or pedon scale (Figure 1) do not address the dynamics of soil hydro-mechanical processes and their interactions (Vereecken et al., 2016). For instance, to better characterize soil water retention curve in a nonrigid soil with dynamic changes in pore structure and solid phase (Gerke et al., 2022), it becomes essential to account for concurrent volume variations (e.g., Horn, 2004; Rainer et al., 2014; Stange & Horn, 2005). In structured soils with biopores, characterized by finer-textured and organic-rich coatings, this dynamic behavior differs locally from that of the matrix (e.g., Leue et al., 2021) such that coated biopores increase the overall hydro-structural stability (Barbosa & Gerke, 2023; Schäffer et al., 2013) of the soil in the vicinity of biopores by reducing the local porosity (Vetterlein et al., 2020) and increasing the particle binding forces due to the presence of exudates (Schäffer

Core Ideas

- Constant hydraulic properties led to prediction errors in nonrigid soil structures.
- Upscaling pore-scale simulations quantified shrinkage and water retention curves.
- Coated biopore surfaces increase soil horizontal hydro-structural stability.

et al., 2013). The intra- and inter-aggregate pores contribute to the complexity due to shrink-swell processes during wetting and drying cycles (Braudeau et al., 2004) and pose additional challenges to prediction and modeling of water and solute movement in soil (Garnier et al., 1997).

The relationship between specific volume and gravimetric water content, denoted as the soil shrinkage curve, generally presents four subdomains defined as (i) structural shrinkage, (ii) proportional shrinkage, (iii) residual shrinkage, and (iv) zero shrinkage (Peng & Horn, 2005). However, the number of subdomains is strictly related to the initial structural condition of the soil (Peng & Horn, 2013). These authors found that structural and proportional shrinkage phases are specific to organic-rich soils that show a well-aggregated soil structure

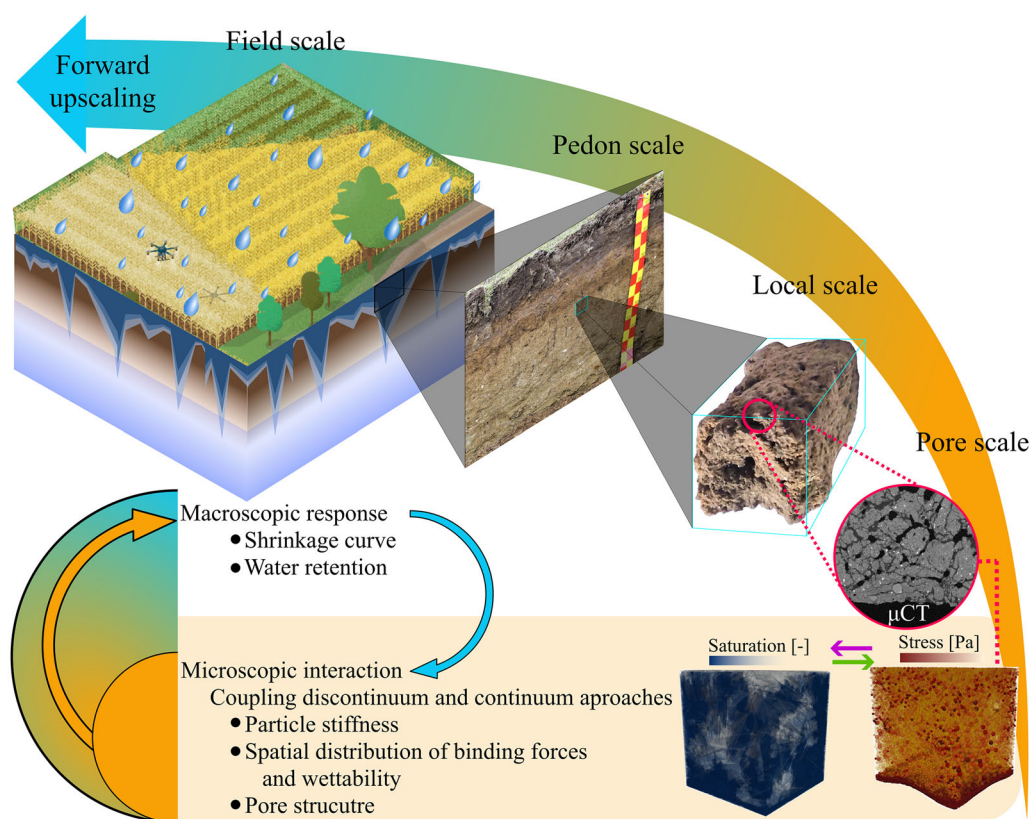


FIGURE 1 Upscaling the influence of microscopic parameters in hydro-mechanical pore-scale modeling (right) on mass exchange processes within the biopore-matrix at the local scale (center), and their effect on shrinkage and hydraulic functions at the pedon and field scales (left).

(Boyle et al., 1989), while soil shrinkage curves without the structural shrinkage phase are derived from highly compacted soils or from aggregates without inter-aggregate pores. Given the sigmoidal shape of the soil shrinkage curve, the exponential model (Braudeau, 1988) is a more flexible approach to fit the shrinkage curve for a large variety of soil structures. It is important to describe soils under different treatments since the land use plays a crucial role on physical properties, resulting in a wide range of shrinkage curve subdomains. The unique feature of such an approach is that it splits the shrinkage of a structured soil into the volumetric decrease of two different components of the pore space on drying: plasma porosity (i.e., intra-aggregate porosity) and structural porosity (inter-aggregate porosity) (Schäffer et al., 2013). The enlargement and propagation of cracks during the drainage process (Bottinelli et al., 2016) may be one of the mechanical responses of the effect of fluid dynamics on soil solid phase (Bishop, 1960; Yuan et al., 2018) in addition to soil bulk shrinkage. The cracks develop due to the greater retraction of colloidal-sized particles (i.e., micro porosity; Schäffer et al., 2013) than the soil bulk volume shrinkage (Bottinelli et al., 2016) leading to the formation of soil desiccation crack systems (Tang et al., 2021).

The formation and evolution of such cracks are dependent on the bond strength of organo-mineral associations (Tisdall & Oades, 1982), which control the fracture surface (Barbosa & Gerke, 2022; Barbosa, Gerke, Munkholm, et al., 2022) and consequently, mass exchange processes across the macropore surface during preferential flow events (Jarvis et al., 2016; Nimmo, 2021). Therefore, the dynamics of soil hydro-structural stability (Schäffer et al., 2008) is critical in the formation and maintenance of primary preferential and secondary pathways for water flow (van der Linden et al., 2019). At pore scale, alterations of organo-mineral association in worm casts and mucus and root exudates (Capowiez et al., 2021) increase the cohesive forces of the surrounding soil particles (Czarnes et al., 2000; Peng et al., 2011). Simulation results (Barbosa, Gerke, & Gerke, 2022) showed that higher coating cohesion (i.e., increased particle bonding force by organo-mineral components) increased the stiffness of the entire biopore structure, suggesting increased mechanical stability of the biopore, as also observed by Schäffer et al. (2008). Furthermore, the organo-mineral coating locally modifies the wettability or sorption properties (Ellerbrock & Gerke, 2013), which may affect the water-holding capacity of the biopore surface (Vetterlein et al., 2020). Organo-mineral components can undergo hydrophobic transformation upon drying, resulting in water repellence within the rhizosphere (Zarebanadkouki et al., 2016).

Although macropores have been extensively studied from the perspective of preferential flow paths (e.g., Coppola et al., 2015; Jarvis et al., 2016; T. Vogel et al., 2010), little is known about the effects of organo-mineral coatings on hydro-

mechanical dynamics of mass exchange processes (Haas et al., 2020) and local non-equilibrium conditions (H. J. Vogel et al., 2023). Gerke and Köhne (2002) used an experimental approach to determine hydraulic properties and estimate mass transfer parameters by comparing infiltration in coated versus uncoated soil clods. These authors and Fér and Kodešová (2012) found that clay-organic coating reduced the hydraulic conductivity considerably as compared to uncoated soil clods. However, neither the mechanical properties nor the structural dynamics of the soil were considered.

To this end, coupling numerical approaches to solve the mechanical complexity of granular and discontinuous materials with fluid dynamics methods at pore-scale level (Figure 1) could shed light on the dynamic interaction between soil deformation, crack formation and propagation, and macroscale hydraulic properties (Barbosa & Gerke, 2023). The quantification of mechanical features at pore scale has been regarded as fundamental to improve our understanding of the structural dynamics by linking it to macroscopic fluid and solute transport behavior (Costanza-Robinson et al., 2011). Recently, a three-dimensional (3D) numerical model coupling a two-phase discrete element method and pore finite volumes has been introduced to simulate hydro-mechanical responses (Yuan & Chareyre, 2017). This technique has been shown to allow investigations of mechanical stresses on the solid phase of complex granular materials, given the balance of forces exerted on wet and non-wet interfaces due to capillary pressure of each pore (i.e., according to the Mayer–Stowe–Princen method), during the transition from the single-phase (water saturated) to the double-phase (air–water) flow system, in the form of pendular bridges (Yuan & Chareyre, 2022).

The pore-scale simulation approach may help to better understand the dynamics of local hydro-mechanic interactions; however, it is crucial to elucidate how the mechanisms influence pedon or field-scale parameters (Figure 1). To effectively parameterize models using pore-scale modeling, a relationship between micro and macro parameters was established (Barbosa & Gerke, 2023). Upscaling techniques (e.g., Vereecken et al., 2007) require to incorporate data from smaller scales (Figure 1), but the impact of soil heterogeneities is often limited (Grayson & Blöschl, 2000; Harter & Hopmans, 2004).

Our hypothesis was that by scaling up pore-scale drainage simulations of the biopore-matrix interface, we may assess how the mechanical properties of the coating material impact the structural dynamics of soil shrinkage. This, in turn, may allow us to quantify the resulting effects on hydraulic parameters during the drying process. For this, we assumed that these mechanical stresses will lead to dynamic changes in soil hydraulic parameters. Here, we aimed at investigating the effect of micro parameters derived from pore structure analysis on macro parameters observed in each subdomain as

characterized by the soil shrinkage curve. We assumed that the properly parameterized micro parameters (i.e., particle stiffness, bond strength, and inter-aggregate pore volume) provide relevant dynamic pore structural information to predict the macroscopic behavior of the soil in terms of hydraulic functions for nonrigid soils. The pore-scale simulations should help identifying interactions and dynamic changes within the pore structure. Model results should demonstrate dynamics in local water retention for soils with biopores during drying. And improved macro parameters derived from the retention curve should help to enhance the prediction of the unsaturated soil hydraulic conductivity.

2 | MATERIALS AND METHODS

2.1 | Soil sampling and characterization

Samples were collected from the Bt horizon of an eroded Luvisol (e.haLV) (Rieckh et al., 2012). The e.haLV soil profile is located in the hummocky arable soil landscape of the Uckermark region in northeastern Germany (53°23' N, 13°47' E). This area is characterized by an average annual precipitation of 489 mm and an annual mean air temperature of 8.6°C, recorded at the Dedelow Experimental Field Station of the Leibniz Centre for Agricultural Landscape Research (ZALF), Müncheberg.

After excavating the surface of Bt horizon, a rectangular part of this surface (1 m × 2 m) was cleaned, and the earthworm burrows were identified for counting the number and for diameter measurement. Lateral cuts into the soil profile were then performed to access the earthworm burrow surfaces (Figure 1). Once the burrows were identified, the burrow structure was collected and cubic samples of approximately 10-mm edge length were extracted from it, where one face of the cube presented the coated earthworm burrow (Figure 2). These samples are denoted as “coated samples” hereafter. The same procedure was performed to extract cubic samples adjacent to the burrow, but without the coated surface. These samples are denoted as “matrix samples” hereafter. Both coated and matrix-only samples were stabilized with three turns of PTFE tape along the side of the cube, leaving the top and bottom exposed. The tape prevented sample disruption during handling and evaporation.

The bulk density values of the matrix of 1.65 kg m⁻³ (standard error of 0.15) and of the coating material of 1.84 kg m⁻³ (standard error of 0.15), as reported before (Barbosa & Gerke, 2023), were used to calculate the volumetric water content from the gravimetric water content for the empirical data (from laboratory observations). From previous investigations, the saturated conductivity of the site is 309 cm d⁻¹ (Rieckh et al., 2012). Note that these values are different from the numerical model that is presented later.

2.2 | Drainage experiments

For the drainage experiments, the sand box was divided into two parts: one for the samples used to quantify the gravimetric water at different pressure heads and another to measure soil shrinkage (Figure 2). On the left side (Figure 2), the replicates were placed in the lines (S1–S6) intercalating matrix and coated samples. The columns (from “a” to “e”) are the samples for each pressure head of 0, –1000, –3160, –6310, and –9000 Pa, respectively. Thus, there were three replicates at each pressure head step for matrix and coated samples. This setup was replicated three times, resulting in a total of 90 samples (i.e., nine replicates for the matrix and the coated samples and five pressure head steps). On the right side of the sand box, three control samples were placed under the measurement field of the optical-laser sensor (scanControl 2700-100; Micro-Epsilon) (Figure 2). A tensiometer was also placed in the sand box to track pressure head over time. The frequency of acquisition for the optical-laser sensor was 10 Hz, and the tensiometer acquisition was 0.2 Hz.

The drainage experiments were designed to promote water movement from the soil matrix through the coated burrow surface; thus, the samples were placed in the sand box with the coated burrow in contact with the sand layer. Since the coated samples presented a curved surface due to the earthworm burrow, the sand box surface had to be prepared with a curved shape in order to guarantee the total contact area between the coated surface and the sand (Figure 2). The maximum height of this curved shape was approximately 2 mm. The matrix samples were placed in the sand layer over a flat surface also 2 mm high to ensure both samples were at the same initial level. In order to minimize the anisotropic effect on the results, the same sample orientation as collected from the field was maintained in parallel for matrix and coated samples.

The experimental setup involved assembling the grid from Figure 2 in a sand box with a pore pressure of –9500 Pa. The water level in the sand box was raised to 0 Pa over 120 s, causing the samples to swell and the optical laser sensor to measure the swelling. Once the swelling stabilized (ca. 30 min), column “a” was removed to measure the saturated weight, followed by oven-drying for 24 h to determine the dry weight and calculate the gravimetric water content. The water level was then lowered to –1000 Pa, and when the shrinking stabilized, column “b” was extracted for gravimetric water measurement. This process was repeated for columns “c,” “d,” and “e” with corresponding pressures.

2.3 | Discrete element model (DEM) of the soil structure

The solid phase of the soil matrix was formed by depositing spherical particles of different sizes in a cuboid packing

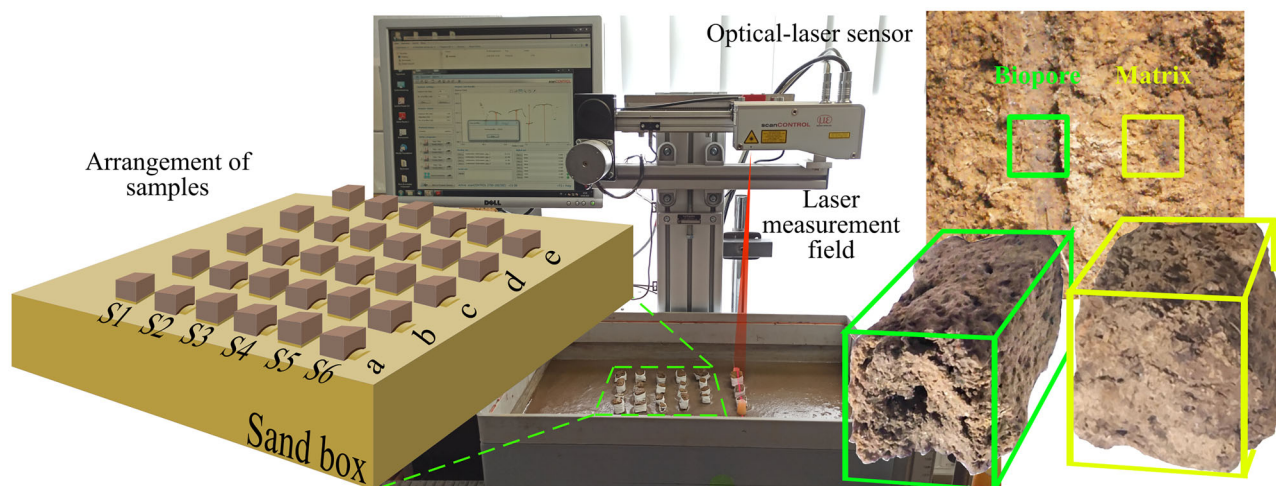


FIGURE 2 Experimental configuration of sand box. Left: Arrangement of samples to quantify the amount of water at different pressure heads. S1–S6 are the replicates intercalating matrix and coated samples. The letters a, b, c, d, and e are the samples in the pressure head of 0, 1000, 3160, 6310, and 9000 Pa, respectively. Right: Detail of the surface of a coated earthworm burrow observed in the soil profile next to the matrix sample and isometric views of a coated and matrix samples, cubes of approx. 10 mm edge.

measuring $10 \text{ mm} \times 10 \text{ mm} \times 10 \text{ mm}$. The larger 1-mm-diameter particles were replaced by an aggregation of 3600 spherical particles with a diameter of 0.052 mm, while the smaller 0.4-mm-diameter particles were replaced by an aggregation of 1260 spherical particles with a diameter of 0.03 mm. The method for creating aggregates is shown in Barbosa et al. (2020). The proportion of large and small aggregates in the packing was determined based on the sand and clay content (549 and 185 g kg^{-1} , respectively) of the Bt-horizon in a e.haLV soil profile (Rieckh et al., 2012).

Once the packing was assembled with the spherical aggregates, it was compressed to achieve a porosity similar to that of the measured soil matrix. Due to computational limitations, a smaller cuboid packing of $5 \times 5 \times 5 \text{ mm}^3$ was extracted from the center of the original one for drainage simulations. This smaller packing contained approximately 25,000 particles with a diameter of 0.052 mm and 54,000 particles with a diameter of 0.03 mm. With this method, we were able to create a soil structure with pores inside aggregates (intra aggregate pores) and pores among aggregates (inter-aggregate pores) (c.f. Barbosa & Gerke, 2023).

A similar procedure was followed for the creation of the matrix of the coated sample model. For the coating itself, the solid phase structure was created within a random cuboid packing of dimensions $5 \times 0.25 \times 5 \text{ mm}^3$. Approximately 60,000 particles with a uniform diameter of 0.015 mm were created for the coating. The particle packing was compressed to achieve a porosity similar to that of the coating material.

The software YADE was used for DEM modeling, and contact law definitions, scene construction, simulation control, and post-processing of data were based on Python programming language (Šmilauer et al., 2015). The cohesive frictional contact law accounting mainly for cohesive frictional mate-

rial, more likely to reproduce the ductile rupture, was already implemented in YADE (Scholtès & Donzé, 2015). The heterogeneity of particle bond strength within the aggregate was controlled by setting the shape parameter of the Weibull distribution (Caulk, 2020) to 1.5 (Barbosa, Gerke, Munkholm, et al., 2022). The cohesive frictional contact law allowed for the recording of breakage events and quantification of crack plane orientation (Scholtès & Donzé, 2013). A global viscous damping of 0.1 was applied to dissipate kinetic energy in all simulations. The model and results of the aggregate pack density are presented in detail by Barbosa and Gerke (2023).

2.4 | Two-phase pore-scale finite volumes coupled to discrete element—DEM-2PFV

The Regular Triangulation method (Chareyre et al., 2012) was used to obtain regular Delaunay facets and Voronoi vertices at pore scale (Figure 1), which defined the pore bodies and throats within the poly-dispersed sphere-packing (Barbosa & Gerke, 2023). The packing triangulation was performed using the CGAL open-source library (Pion & Teillaud, 2011), resulting in a mesh that accounted for the local pore geometry, such as the radii of the solid phase and the effective entry pore throat area. This mesh defined the finite volume of the pore-network model, and the local pore geometry was used to calculate fluxes between connected tetrahedral elements based on pore pressure gradient (Chareyre et al., 2012).

The drainage process was controlled by gradually decreasing the capillary pressure, which represents the pressure difference between the air and water phases. The initial state of the packing was assumed to be water saturated. The capillary pressure difference was achieved by setting the air

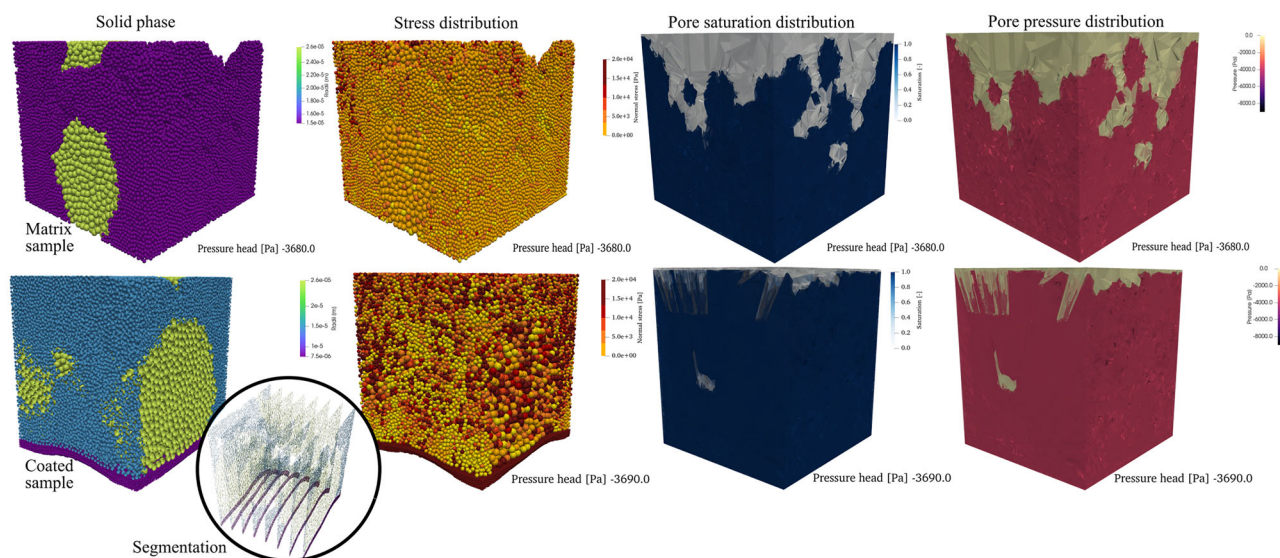


FIGURE 3 Discrete element model coupled with two-phase pore-scale finite volume of matrix (top) and coated samples (bottom). Solid phase shows the distribution of aggregate and particles sizes and the effect of pore pressure on stress and saturation distribution. Detail of segmentation of three-dimensional (3D) samples into two-dimensional (2D) slices for imaging processing and quantification of inter-aggregate pore volume.

pressure at the top of the packing to 0 Pa and imposing an initial water pressure of 0 Pa at the bottom. Drainage commenced by gradually decreasing the water pressure while keeping the air pressure constant. The water pressure was decreased in steps of 50 Pa until a maximum of 10 kPa was reached. After each pressure step at the bottom boundary, the packing was allowed to stabilize and the forces were equilibrated. The mesh of the triangulation was upgraded to account for particle movement before the next pressure step was applied (Figure 3).

In this approach, the particles were considered perfectly wettable by water, and viscous effects in the fluids were considered negligible. The pore water drainage started when the pore water pressure exceeded the entry capillary pressure determined by the pore throat geometry (Figure 3). The pore-scale process known as Haines jump could trigger a chain reaction, resulting in the drainage of multiple pores simultaneously (Yuan & Chareyre, 2022). Additionally, air or water could become disconnected from their reservoirs (at the top or bottom, respectively), causing both phases to become trapped within each other. The air–water interfacial tension and the geometry of particles in contact were utilized to calculate the maximum value of the entry capillary pressure and the distribution of hydrostatic forces. For a comprehensive explanation of the force equations, please refer to Yuan and Chareyre (2017).

2.5 | Parameterization approach

The parameterization process has been described in detail before (Barbosa & Gerke, 2023). Briefly, the 250 drainage

simulations using the matrix structure were carried out by randomly varying the micro parameters (i.e., particle stiffness and bond strength) to determine strain versus pressure head relations. These macro parameters were quantified for each combination of micro parameters using multi-output regression. Machine learning was employed to find the dependency network between microscopic (independent variable) and macroscopic (dependent variable) parameters. The optimized micro parameters (i.e., particle stiffness and bond strength) obtained for the soil matrix were then used for the coated biopore sample, and the procedure starting with drainage simulations for coated matrix samples was carried out by again randomly varying values of the particle stiffness and bond strength for the coating material.

Note that the machine learning algorithm was used to fit the numerical results (experimental data from simulation scenarios) to macro parameters obtained experimentally by means of a nonlinear optimization process. Since the soil and the model used here were exactly the same as in the previous study (Barbosa & Gerke, 2023), the parameterized values of particle stiffness and bond strength were the same as in the previous study. Therefore, the particle stiffness of the matrix sample was 6.7 MPa and that of the coating material was 13.6 MPa. The bond strength was 870 Pa for the matrix and 500 MPa for the coating material.

2.6 | Upscaling hydraulic parameters

The soil hydraulic parameters of empirical (obtained from water retention in sand box experiments) and experimental data (from simulation scenarios) were described using

unimodal (subscript $i = 1$) and bimodal (subscript $i = 2$) van Genuchten (vG and bi-vG) water retention model (Durner, 1994; Mualem, 1976; van Genuchten, 1980) as:

$$S(h) = \sum_{i=1}^2 \left(w_i [1 + (\alpha_i |h|)^{n_i}]^{-m_i} \right); m_i = 1 - 1/n_i \quad (1)$$

where h pressure head (Pa), α [L^{-1}], n , and m are empirical coefficients, here obtained from by fitting, w_1 and w_2 are the volume fractions of the pore domains ($1 = w_1 + w_2$), and S is effective saturation calculated from the volumetric water content (θ) ($cm^3 cm^{-3}$) as:

$$S = (\theta - \theta_r) / (\theta_s - \theta_r) \quad (2)$$

where θ_r denotes the residual and θ_s the saturated water content parameter ($cm^3 cm^{-3}$).

The unsaturated hydraulic conductivity was described as:

$$K(S) = K_s S^l \left(\left[1 - \left(1 - S^{\frac{1}{m}} \right)^m \right] \right)^2, l = 0.5 \quad (3)$$

where θ denotes water content ($cm^3 cm^{-3}$), K_s [LT^{-1}] is the saturated hydraulic conductivity, and l is the pore-connectivity coefficient (dimensionless).

For matrix and coated sample analysis, hydraulic parameters of empirical and experimental data were described using the bimodal-porosity retention functions (Equation 1) and the closed-form expression of the unsaturated hydraulic conductivity function (Priesack & Durner, 2006) as:

$$K(S) = K_s (w_1 S_1 + w_2 S_2)^l \frac{\left(w_1 \alpha_1 \left[1 - \left(1 - S_1^{\frac{1}{m_1}} \right)^{m_1} \right] + w_2 \alpha_2 \left[1 - \left(1 - S_2^{\frac{1}{m_2}} \right)^{m_2} \right] \right)^2}{(w_1 \alpha_1 + w_2 \alpha_2)^2} \quad (4)$$

The Python library unsatfit (Seki et al., 2022) was used to fit parameters of vG and bi-vG soil hydraulic functions for both, empirical and experimental data.

In addition to calculating the hydraulic conductivity and the effective saturation based on hydraulic parameters, they were also determined by numerical modeling, called experimental data hereafter. Unlike the calculated value of unsaturated hydraulic conductivity ($K(S)$), which is based on a constant value of saturated hydraulic conductivity (K_s), the hydraulic conductivity obtained from experimental data incorporated the dynamic variations in permeability resulting from structural changes during the drying process.

2.7 | Shrinkage curve model

The bulk soil shrinkage curve, here described in terms of specific volume versus gravimetric water content, was

TABLE 1 System of equations from exponential model with the 12 model parameters (gravimetric water content, W [$kg kg^{-1}$], and specific bulk volume, V [$cm^3 g^{-1}$]) of the six transition points (W_{IS} , W_{IL} , W_{MS} , W_{ML} , W_{AE} , W_{SL} , V_{IS} , V_{IL} , V_{MS} , V_{ML} , V_{AE} , and V_{SL}) and slopes of the linear shrinkage domains (K_{ip} , K_{st} , K_{bs} , and K_{re}).

Linear domain	Slopes of linear domains
Interpedal shrinkage domain	$K_{ip} = \frac{(V_{IS}-V_{IL})(e-1)}{W_{IS}-W_{IL}} - K_{st}(e-2)$
Structural shrinkage domain	$K_{st} = \frac{(V_{MS}-V_{ML})(e-1)}{W_{MS}-W_{ML}} - K_{bs}(e-2)$
Basic shrinkage domain	$K_{bs} = \frac{-K_{st}+(e-1)(V_{MS}-V_{ML})/(W_{MS}-W_{ML})}{e-2}$
Residual shrinkage domain	$K_{re} = \frac{(V_{AE}-V_{SL})(e-1)}{W_{AE}-W_{SL}} - K_{bs}(e-2)$

Note: e is the Euler's number.

determined for coated and matrix-only samples from water saturation, pore volumes, and strain obtained from pore-scale simulations of drainage. The simulations carried out on parameterized provided values of the sample saturation, the average water flow velocity, the strain, the porosity, and the crack surface area as a function of the pressure head. These values were used to fit the exponential shrinkage curve model (Braudeau et al., 1999). The system of model equations was solved by nonlinear least square optimization using the SciPy optimize function (Virtanen et al., 2020). The following transition points were calculated: (i) interpedal shrinkage (IS), (ii) the interpedal shrinkage limit (IL), (iii) maximum swelling (MS), (iv) macroporosity limit (ML), (v) air entry (AE), and (vi) the shrinkage limit (SL). The slopes of the four linear shrinkage subdomains were assumed (Braudeau et al., 1999) to correspond with interpedal shrinkage (K_{ip}), structural shrinkage (K_{st}), basic shrinkage (K_{bs}), and residual shrinkage (K_{re}). These slopes of each linear subdomain depended on the bulk sample volume (V) and the gravimetric water content (W) of each transition point (Table 1). More details about these shrinkage curve sections can be found in Schäffer et al. (2013).

2.8 | Macropore volume variation

The DEM-2PFV simulations were post-processed using Paraview, and the 3D samples were segmented into two-dimensional (2D) slices using video creation and image export (Figure 3). The voxel resolution of the image corresponded to 3 μm . The samples were then binarized using FIJI and processed using morphological segmentation (Legland et al., 2016). The parameters for the segmentation algorithm included gradient radius 1, watershed segmentation tolerance 20, and displayed as catchment basins. Watershed segmentation was done using local minima of the gradient of the image

TABLE 2 Parameters of the soil hydraulic functions obtained by fitting experimental (Exp.) and pore-scale simulated (Sim.) data, for nonrigid coated and matrix samples and rigid matrix.

	Sample		K_s (m s ⁻¹)	w_1	α_1 (cm ⁻¹)	n_1	α_2 (cm ⁻¹)	n_2	m_1	RMSE
Exp.	Coated	bi-vG		0.32	0.18	5.34	0.008	5.21	0.69	2E-6
		vG		n.a.	0.17	1.59	n.a.	n.a.	0.32	2E-2
	Matrix	bi-vG		0.24	0.084	6.46	0.0084	5.57	0.84	2E-6
		vG		n.a.	0.93	1.12	n.a.	n.a.	0.11	3E-2
Sim.	Coated	bi-vG	0.0094	0.27	0.057	17.27	0.01	2.69	0.96	9E-3
		vG		n.a.	0.05	1.72	n.a.	n.a.	0.42	2E-2
	Matrix	bi-vG	0.6615	0.42	0.083	2.69	0.0079	5.87	0.63	3E-3
		vG		n.a.	0.098	1.73	n.a.	n.a.	0.42	9E-3
	Rigid matrix	bi-vG		0.38	0.069	7.11	0.012	1.48	0.86	6E-3
		vG		n.a.	0.069	2.90	n.a.	n.a.	0.66	1E-2

Abbreviation: n.a., not available; RMSE, root mean square error; K_s , hydraulic conductivity.

to separate voids from the solid phase of the soil structure for all 2D slices of the region of interest (Barbosa, Gerke, & Gerke, 2022). The 3D inter-aggregate pore structure was recreated from this segmentation. This macropore volume was calculated for the pressure heads of 0, -1000, -2000, -3000, -4000, -5000, -7500, and -10,000 Pa, and the deviation from the initial state at a pressure head of 0 Pa was related to the gravimetric water content.

3 | RESULTS

3.1 | Parameterized drainage simulations and empirical data

The water retention of the empirical and experimental data for both matrix and coated samples displayed best fitting with the bimodal (bi-vG), as shown by the smaller root mean square error (RMSE) in Table 2 (c.f. Figure S1). While the rigid matrix displayed a slightly smaller α_1 (higher air entry point), the nonrigid matrix displayed a significantly smaller α_2 (Figure 4). The values of α between simulation and experimental data were relatively close; however, a significantly difference between simulated and experimental α_1 of the coated sample was observed. From the simulations, the first air entry point for the coated sample was slightly higher compared to the matrix, which was reversed for the second air entry point. The main discrepancy between α_2 measured (9500 Pa) and calculated (12,000 Pa) from experimental data was for the matrix (Figure 4, left side). Within the bimodal fittings, the simulations yielded higher n_1 values for the rigid matrix in comparison to the nonrigid one and for the coated sample compared to the experimental value (Table 2). This scenario, however, was the opposite for n_2 values. At the beginning of the numerical simulations, the K_s values were determined to be 0.6615 m s⁻¹ for the matrix sample and

0.0094 m s⁻¹ for the coated sample (Table 2). These values were used for the calculation of empirical and experimental hydraulic conductivity. The calculated hydraulic conductivity (Figure 4, right side) obtained from the parameters of the simulated and experimental water retention (Table 2) displayed RMSE of 1.4E-3 for the coated sample and 1.6E-1 for the matrix. The RMSE between the predicted hydraulic conductivity and the measured experimental data was 1.2E-3 for the coated sample and 2.1E-1 for the matrix. The effective saturation (S) obtained by utilizing the bimodal parameters from Table 2 in simulations of the coated sample exhibited an average relative error of 0.7% as compared to the measured value (Figure 5). Conversely, the average relative error between the calculated and measured effective saturation for the matrix was 3%. The progressive lag between the curves becomes increasingly noticeable in the shrinkage subdomain IX of the coated sample and in subdomain III of the matrix.

Simulations of drainage in matrix display a similar peak in the frequency of pore throat radii of P_1 , P_2 , and P_3 in the pressure head of 520 Pa. From this point on, the main drainage was observed for pores of sizes P_2 and P_3 , increasing the frequency approximately 100 times from 520 to 9520 Pa. Simulations of drainage in coated sample displayed peaks of frequency for pore throat radii of P_2 and P_3 in the pressure head of 1560 Pa. From this point up to 1890 Pa, the main drainage was observed in larger pores (between P_1 and P_2). Between 1890 and 9510 Pa, the main drainage was once again observed in smaller pores (P_2 , P_3 , and P_4), also increasing the frequency approximately 100 times from 1560 to 9510 Pa (Figure 6).

3.2 | Shrinkage curves and parameters

The structural shrinkage was divided into seven subdomains (four linear and three nonlinear) for the matrix drainage simulations, while the coated sample presented nine subdomains

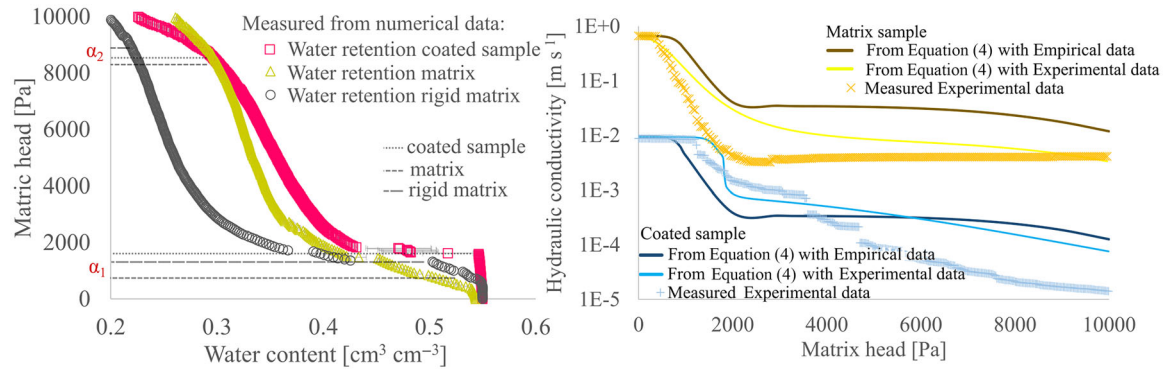


FIGURE 4 Hydraulic parameters determined from numerical data and drainage experiments. Left: Water retention curves obtained from numerical simulations, where α_1 and α_2 are the measured values of the inverse air-entry pressures. Right: Hydraulic conductivity for the coated and matrix samples calculated from bi-vG model (Equation 4) for empirical and experimental data (from laboratory observations and from simulation scenarios, respectively), as well as the measured hydraulic conductivity from simulations (measured experimental data).

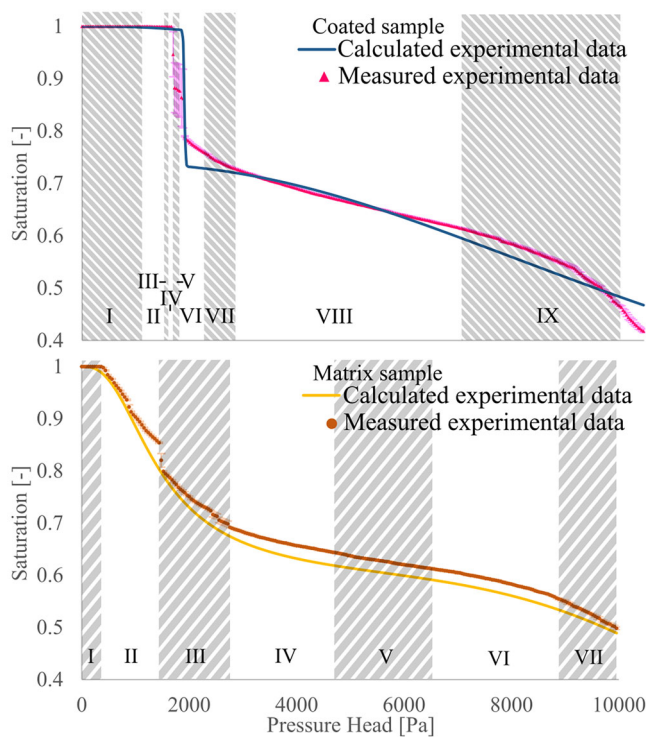


FIGURE 5 Effective saturation measured from empirical data and calculated from bi-vG model (Equation 1) for coated (top) and matrix samples (bottom). The hatched lines and Roman numerals identify the subdomains of the shrinkage curve of each sample.

(five linear and four nonlinear) (Figure 7). Therefore, the shrinkage of the coated sample presented two more transition points (W_{IS-2} , W_{IS-1} , V_{IS-2} , and V_{IS-1}) and one more linear subdomain (K_{st-1}) compared to the matrix. The exponential model (Braudeau, 1988) was able to fit the data from both samples, and the parameters of the shrinkage curve are presented in Table 3.

3.3 | Pore structure dynamics

The inter-aggregate pore volume in the first structural shrinkage subdomain of the coated sample (K_{st-1}) increased by approximately 80% (gravimetric water content of 0.4387 kg kg⁻¹) (Figure 7). Conversely, in the matrix sample, which exhibited initial interpedal shrinkage (K_{ip}), the inter-aggregate pore volume decreased by 57% at the same gravimetric water content level. A substantial decrease in the pore volume between aggregates, amounting to a total variation of 105%, was observed during the transition from structural shrinkage to basic shrinkage in subdomains VI–VIII of the shrinkage curve of the coated sample (Figure 7), dynamically impacting water retention and hydraulic conductivity (Figure 4). At the end of the structural shrinkage (subdomain IV), the matrix exhibited a peak variation in pore volume between aggregates (relative measure of pore volume), reaching approximately 15% (Figure 7). Additionally, the coated sample displayed a 22% increase in pore volume at the end of the basic shrinkage (subdomain VIII).

While the coated sample exhibited a higher degree of variation in inter-aggregate pore volume, the matrix displayed lower hydrostatic stability (greater slope in phases I and II, Figure 7), leading to a more pronounced reduction in the sample-specific volume during the drying process. This soil shrinkage behavior is macroscopically included in the 3D graphs of the soil water retention curve as volumetric water content versus specific volume and pressure head (Figure 8, right) and by the calculated hydraulic conductivity versus specific volume and water content (Figure 8, left); these graphs describe the characteristically different curved shapes for coated and matrix samples for the simulation data that included hydro-mechanic structural dynamics (Figure 8).

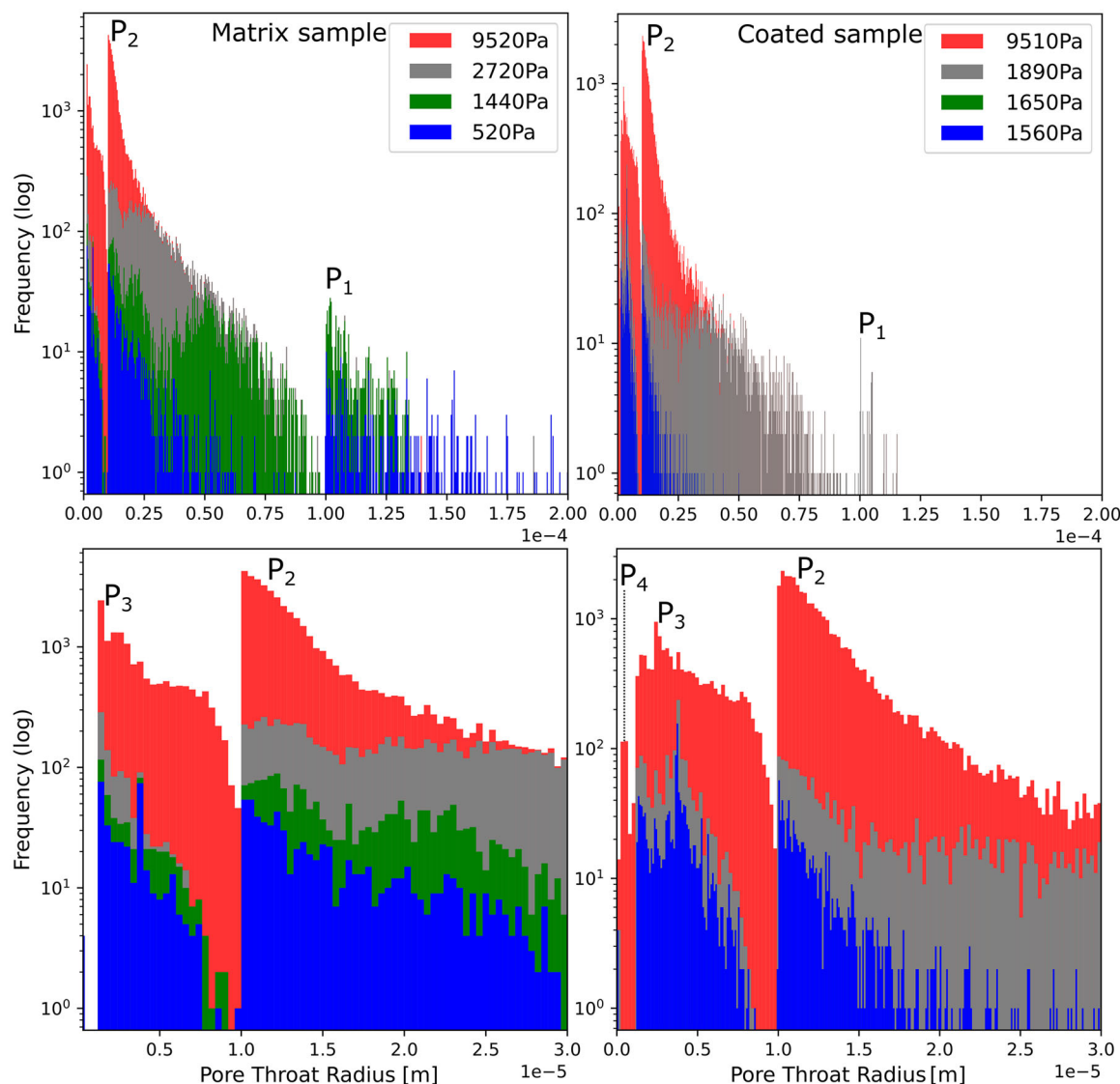


FIGURE 6 Distribution of pore throat radius drainage for the matrix (left) and coated sample (right) for different pressure head values. The three main pore sizes common to both samples are as follows: (i) the inter-aggregate pore throat ($P_1 > 1e-4$ m), (ii) pore throat within the 1 mm aggregate ($1e-4 < P_2 < 1e-5$ m), and (iii) pore throat within the 0.4 mm aggregate ($1e-5 < P_3 < 1.3e-6$ m). Additionally, the coated sample presented a pore throat within the coating material ($P_4 < 1.3e-6$ m). The zoom in on the histograms between pores P_2 and P_4 provides more drainage details (bottom).

4 | DISCUSSION

4.1 | Soil structure dynamics during drainage

The shrinkage curve (Figure 7) for the matrix contains the subdomains Interpedal (I–II), Structural (III–IV), Basic (V–VI), and Residual (VII) shrinkage, typical of a structured soil (Braudeau et al., 1999; Schäffer et al., 2013). The coated sample presented interpedal to residual subdomains from III to IX, and beyond interpedal, it presented two more subdomains (I–II) (Table 3). The effect of plastic deformation observed in subdomains I–II of the matrix shrinkage curve

and in subdomains VII–VIII of the coated sample (Figure 7) is linked to the abrupt collapse of preexisting macropores, as shown by the reduction of inter-aggregate pore volume. The closure of macropores significantly decreases the specific volume of the sample, presenting higher slopes in the linear subdomains (subdomain I of the matrix sample) (Table 3). However, smaller values of the linear slope of the subdomains at the beginning of the shrinkage (subdomain I of the coated sample) are followed by a significant increase in the volume of macropores (Figure 7). The shrinkage of smaller pores may cause the creation or expansion of preexisting cracks (Bottinelli et al., 2016; Peng & Horn, 2013). Examining the subdomains I–VI of the coated sample in Figure 7 reveals

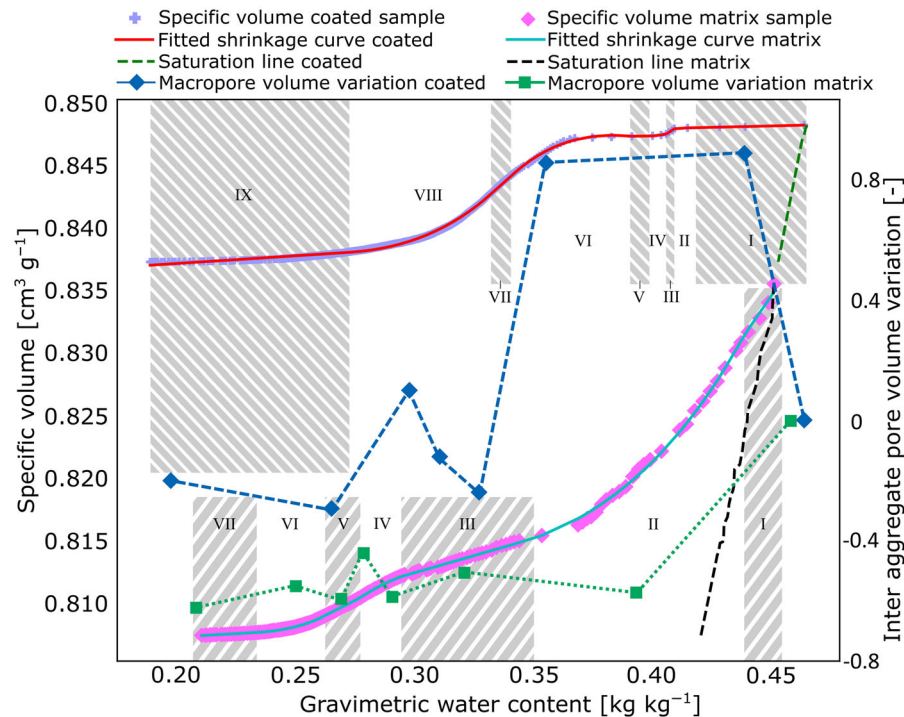


FIGURE 7 Soil structural shrinkage and inter-aggregate pore volume variation during drainage simulations of matrix and coated sample. Fitted exponential model subdivided into linear and nonlinear shrinkage subdomains I–VII for the matrix and I–IX for the coated sample, where odd numbers show the linear domains and even numbers show the nonlinear domains.

TABLE 3 Parameters of exponential model and shrinkage domains obtained from shrinkage curves of numerical simulations.

Sample	W_{IS-2}	W_{IS-1}	W_{IS-0}	W_{IL}	W_{MS}	W_{ML}	W_{AE}	W_{SL}
Matrix	n.a.	n.a.	0.4383	0.3501	0.2944	0.2773	0.2625	0.2338
Coated	0.410	0.409	0.406	0.392	0.39	0.341	0.332	0.273
	V_{IS-2}	V_{IS-1}	V_{IS-0}	V_{IL}	V_{MS}	V_{ML}	V_{AE}	V_{SL}
Matrix	n.a.	n.a.	0.8314	0.8152	0.8121	0.8105	0.8090	0.8077
Coated	0.84798	0.84793	0.8475	0.847331	0.847332	0.8442	0.8427	0.838
Linear domains								
	K_{st-1}	K_{ip}	K_{st-0}	K_{bs}	K_{re}			
Matrix	n.a.	0.2767	0.0545	0.0971	0.0107			
Matrix shrinkage domains	n.a.	I	III	V	VII			
Coated	0.0045	0.125	-0.0198	0.170	0.0123			
Coated shrinkage domains	I	III	V	VII	IX			

Abbreviation: n.a., not available.

that drainage primarily affects the intra-aggregate pores (P_2 , P_3 , and P_4), as shown by the blue and green histograms of pore-scale results (Figure 6). Consequently, this explains the increase in macropore volume observed for the coated samples (Figure 7). Conversely, the subdomain I–II of the matrix shrinkage is a result of the drainage of both inter- (P_1) and intra-aggregate pores (P_2 and P_3), also depicted by the blue and green histograms in Figure 6 (i.e., here drainage is causing a collapse of macropores).

Braudeau et al. (1999) attributed the higher shrinkage in subdomain I (i.e., interpedal shrinkage) to a relatively weak

cohesion (i.e., smaller rigidity) of the soil structure. Our drainage simulations (Figure 7) of the matrix sample confirm this statement because. The effective stress (Bishop, 1960) breaks the bond strength between particles more easily in the matrix sample than in the coated sample. This leads to a larger relative particle displacement and aggregate breakage in the matrix (Figure 7), which can be interpreted as higher plastic deformation (Asaf et al., 2007; Cil & Alshibli, 2012). In the coated sample, the higher cohesion of the coating particles buffers the stress distribution within aggregates in the matrix, reducing the bulk plastic deformation (Balakrishnan et al.,

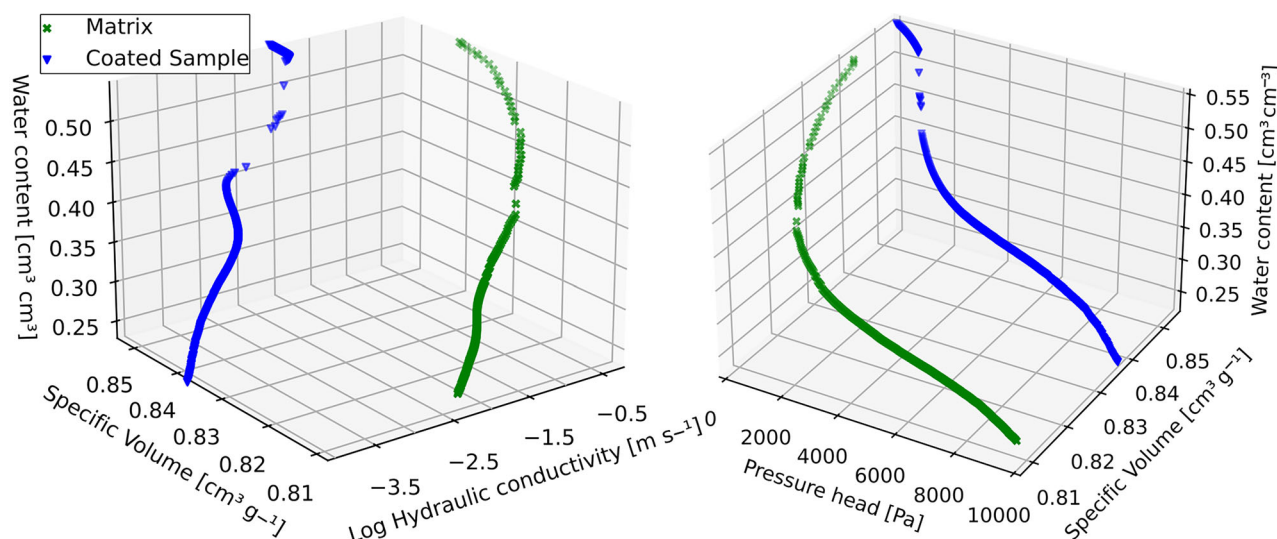


FIGURE 8 Integration of soil water retention and shrinkage curves. Left: The relationship between volumetric water content, calculated hydraulic conductivity and specific volume. Right: The relationship between water content and specific volume alongside the corresponding pressure head.

2010). Therefore, the set of bonding forces between particles and aggregates, along with the rigidity of the particles, determines the rigidity of the overall soil structure (Barbosa & Gerke, 2023), which can vary depending on the drainage of structural or plasma pores, as demonstrated here in the form of the shrinkage curves (Figure 7).

The shrinkage curves (Figure 7) reveal smaller slopes in the linear subdomains, indicating, according to Boivin et al. (2006), a higher hydro-structural stability for the coated sample. Hydro-structural stability refers to the ability of the soil to withstand internal stresses resulting from decreasing water potentials during the drying process (Schäffer et al., 2013). Our findings (Figure 7) demonstrate that the matrix structure becomes stiffer in subdomain III (Table 3), leading to a reduction in shrinkage that impacts the water retention curve pattern (Figure 8). This is due to the two-way coupling feature of the model (Barbosa & Gerke, 2023). This reduction occurs as the drainage shifts from inter- to intra-aggregate pores (P_2 and P_3), as depicted in the gray histogram (Figure 6). Although this hardening behavior is less pronounced in the coated sample, it does occur toward the end of subdomain VII and the beginning of subdomain VIII, known as basic and residual shrinkage (Schäffer et al., 2013). Notably, the coated sample exhibits the highest shrinkage rate at this point, coinciding with a significant decrease in macropore volume (Figure 7). While our numerical results do not support the notion that slopes greater than 1.0 indicate soil structure collapse or that slopes smaller than 1.0 signify structural pore enlargement (Peng & Horn, 2013), they do indicate that a smaller slope in the linear subdomain of the shrinkage curve corresponds to the creation or enlargement of macropores (i.e., cracks) during shrinkage. Conversely, larger slopes suggest structural

collapse (Bottinelli et al., 2016). The specific threshold slope value determining this structural dynamic may not only be an inherent property of the soil type but also includes effects of management practices (Peng & Horn, 2005). Additionally, it is important to acknowledge that the calculated macropore volume variation values obtained from the model may deviate from empirical values. Specifically, the model's absolute values range from approximately -60% to 80% for a volume of interest (VOI) measuring $5 \times 5 \times 5 \text{ mm}^3$. In contrast, Bottinelli et al. (2016), who studied a larger VOI of $30 \times 30 \times 30 \text{ mm}^3$, observed much smaller variations of approximately 4% . Therefore, to accurately quantify the dynamics of the pore network, it is essential to conduct a sensitivity analysis to define a representative elementary volume (Gerke et al., 2022).

Given the importance of accurately characterizing the dynamics of the pore network, incorporating 3D image by using x-ray CT imaging (Figure 1) alongside pore-scale modeling can provide a more realistic representation of the soil structure (Schnepf et al., 2022). Ultimately, such a comprehensive approach can lead to a better understanding of soil hydro-structural behavior shedding light to untangle the importance of the different physical properties in the shrinkage process.

4.2 | Coating effect on soil hydraulic functions

Analysis of the α_1 parameter (Table 2) revealed that the matrix sample had significantly smaller air-entry values of pressure head (1000 Pa) compared to the coated sample (1728 Pa). These values were in agreement with the simulation-measured

data (Figure 4, left side). The higher air-entry point of the coated sample also explains the higher n_1 value obtained from the fitting of bi-vG (Table 2), in agreement with van Genuchten and Nielsen (1985). In addition, at this point, matrix displayed more inter-aggregate pores ($>P_1$, Figure 6) entering air compared to the coated sample, as indicated by the green matrix histogram and gray coated sample histogram in Figure 6. Qian et al. (2022) also found that the first air-entry point was related to the drainage of inter-aggregate pores given the smaller entry capillary pressure of larger pore throats (Yuan & Chareyre, 2022). The second air entry instant calculated for the coated sample (α_2) (Table 2) corresponded to a pressure of approximately 9660 Pa, which was coherent with the measured data from the simulations (Figure 4, left side). The capillary pressure exceeded the entry value given by the pore throat P_2 , P_3 , and the pores from the finer-textured coating material (P_4), as shown by the histogram in Figure 6. At this point, the water retention curves of both samples intersected at approximately $0.3 \text{ cm}^3 \text{ cm}^{-3}$. The calculated and measured α values for the rigid matrix also exhibited good agreement (Table 2 and Figure 4, right side). However, the second air entry instant calculated for the matrix presented a value of 12,000 Pa, which was above the measured value of approximately 9500 Pa (Figure 4, left side). Such a difference observed between measured and calculated values of α_2 for the matrix can be attributed to the lower hydro-structural stability of the matrix sample (indicated by greater shrinkage, as shown in Figure 7) compared to the coated sample. As a result, at higher pressure head values, the structural changes in the matrix sample can impact hydraulic parameters such as the values of α and n , which are assumed to be constant in the bimodal van Genuchten (bi-vG) model. Consequently, this may lead to prediction errors when analyzing nonrigid soil structures (Stange & Horn, 2005).

The initial K_s value, which is comparable with that of a rigid pore structure, was about 70 times higher for the matrix-only sample than for the coated sample (Table 2), which is in agreement with Gerke and Köhne (2002) and Fér and Kodešová (2012), and can be explained by the finer-textured coating material. Therefore, at the beginning of the drainage simulation, the unsaturated hydraulic conductivity calculated and measured for the matrix was higher than the value obtained for the coated sample (Figure 4, right side). However, the lower matrix hydro-structural stability led to a rapid reduction of macropore volume (Figure 7), as discussed in the previous section. Although the K_s value does not provide the absolute empirical value, the qualitative comparison between the empirical and experimental curves (Figure 4) suggests that the K_s value was also varying during the drying process. Similarly to α and n , the dynamic structure could also lead to prediction errors when analyzing nonrigid soil structures (Stange & Horn, 2005), which requires further investigations.

4.3 | Combining soil water retention and shrinkage

The significant disparity observed in water retention simulations between rigid and nonrigid soils (Figure 4, right side) emphasizes the critical significance of accounting for the influence of structural dynamics on drying processes. Although the measured and calculated effective saturation curves from simulated data exhibited a small relative error (Figure 5), mainly at higher pressure heads (shrinkage subdomain IX of the coated sample and from subdomain III of the matrix), the hydraulic conductivity values of rigid soil differed from those of nonrigid soil (Figure 4, right side). This difference could be attributed to the fact that the dynamic structural changes in the soil have a smaller impact on vG α and n parameters, which are utilized in the effective saturation calculation (Equation 1). However, the determination of hydraulic conductivity relies on an initial K_s value that is assumed to remain constant, which is not consistent with the results of the inter-aggregate pore volume variation (Figure 7). This elucidates the larger difference of curve shapes observed between the measured and calculated values obtained from the simulation (Figure 4, right side).

Rainer et al. (2014) have previously observed similar dynamics and emphasized the importance of considering the nonrigidity of soils when trying to describe dynamic processes. The combination of macroscopic parameters related to soil shrinkage and the calculated water retention and unsaturated hydraulic conductivity functions, along with the specific volume and the volumetric water content, allowed for the depiction of distinct 3D-curved shapes (as proposed by Stange & Horn, 2005) that are characteristic for the coated and the matrix-only samples (Figure 8).

However, the question remains as follows: How can current shrinkage curve and soil hydraulic models, such as those proposed by Durner (1994), be adapted to incorporate the mechanical dynamics of the soil? Which mechanical properties play a more significant role in this model reformulation? Some attempts have previously been made (Coppola et al., 2012), but the mechanistic approach to answer these questions remains an open issue. As a more direct solution, one may describe the parameters α and n of the vG model as a function of the void ratio (Stange & Horn, 2005). However, the void ratio is a consequence of micro property interactions between bond strength and particle characteristics that are still unclear. Recent studies by Barbosa, Gerke, and Gerke (2022) and Barbosa and Gerke (2023) have highlighted the importance of micro parameters such as particle bond strength and stiffness obtained from pore-scale models for the dynamics of macropore volume during the drying process.

5 | CONCLUSIONS

In this investigation, we conducted pore-scale drainage simulations of flow in clod samples with and without biopore coatings. The mechanical micro properties of the coating material affected the overall structural dynamics of macroscale soil shrinkage and soil hydraulic properties.

The drainage simulations revealed that lower bond strength among particles in the matrix favored higher shrinkage, while the higher particle stiffness and bond strength of the finer-textured coating material provided greater hydro-mechanical stability for the coated sample. The coupled two-phase pore-scale finite volume discrete element model (DEM-2PFV) simulations allowed to quantify the dynamic formation and collapse of macropores in drying samples across different shrinkage subdomains. Specifically, these simulations confirmed that early drainage of intra-aggregate pores increased the volume of preexisting pores, whereas later drainage of inter-aggregate pores led to the collapse of the soil structure and subsequent volume reduction of preexisting pores. The simulations also confirmed that the extent of soil structure hardening during the drying process depended on the degree of the shift in drainage from inter- to intra-aggregate pores.

Moreover, the simulations characterized a presumed disparity in water retention and hydraulic conductivity curves between rigid and nonrigid soils. While the bimodal model demonstrated the best fit for matrix and coated samples from the empirical and experimental data, the difference between measured and calculated experimental values of α_2 for the matrix-only sample revealed an expected effect of a greater shrinkage as compared to the coated sample. The larger discrepancy between the measured and calculated hydraulic conductivity curve shapes of the numerically determined data confirmed pore structural changes due to hydro-mechanical dynamics during sample drainage not yet considered by dynamically changing the K_s values.

Through upscaling the findings from the pore-scale model DEM-2PFV, we were able to evaluate the influence of the coating material on macro parameters of both soil shrinkage and soil hydraulic parameters. These results provide a step toward mechanistic investigations to adapt current macroscopic soil hydraulic models by incorporating such pore-scale hydro-structural dynamics into hydraulic parameters. The hydro-mechanic soil dynamics is of paramount importance for enhancing the understanding of soil physical processes including the description of macropore-matrix mass exchange, thereby enabling more accurate predictions of preferential flow and transport in structured soils. The future could integrate the numerical pore-scale model with imaging techniques to empirically quantify the structural dynamics of the soil and provide the model with information derived from the pore network.

AUTHOR CONTRIBUTIONS

Luis Alfredo Pires Barbosa: Conceptualization; data curation; formal analysis; investigation; methodology; resources; software; validation; visualization; writing—original draft.
Horst H. Gerke: Conceptualization; project administration; supervision; writing—review and editing.

ACKNOWLEDGMENTS

We gratefully acknowledge the financial support by the German Research Foundation (DFG) under grant GE 990/14-1, entitled: “The exchange flow between preferential flow paths and matrix in soils: from pore to continuum scale with tensors” (FLEXPO). Thanks to Kristian Berger (ZALF) and Annelie Ehrhardt (ZALF) for the technical support in the laboratory.

ORCID

Luis Alfredo Pires Barbosa  <https://orcid.org/0000-0001-9210-8542>

REFERENCES

- Asaf, Z., Rubinstein, D., & Shmulevich, I. (2007). Determination of discrete element model parameters required for soil tillage. *Soil and Tillage Research*, 92, 227–242. <https://doi.org/10.1016/j.still.2006.03.006>
- Balakrishnan, A., Pizette, P., Martin, C. L., Joshi, S. V., & Saha, B. P. (2010). Effect of particle size in aggregated and agglomerated ceramic powders. *Acta Materialia*, 58, 802–812. <https://doi.org/10.1016/j.actamat.2009.09.058>
- Barbosa, L. A. P., & Gerke, H. H. (2022). Structural heterogeneity of soil clods: Correlating Weibull parameters to fracture surface topography. *Geoderma*, 428, 116161. <https://doi.org/10.1016/j.geoderma.2022.116161>
- Barbosa, L. A. P., & Gerke, H. H. (2023). Continuum and discrete element modelling for describing coupled hydro-mechanical effects of earthworm burrow coatings on soil shrinkage. *Geoderma*, 435, 116497. <https://doi.org/10.1016/j.geoderma.2023.116497>
- Barbosa, L. A. P., Gerke, K. M., & Gerke, H. H. (2022). Modelling of soil mechanical stability and hydraulic permeability of the interface between coated biopore and matrix pore regions. *Geoderma*, 410, 115673. <https://doi.org/10.1016/j.geoderma.2021.115673>
- Barbosa, L. A. P., Gerke, K. M., Munkholm, L. J., Keller, T., & Gerke, H. H. (2022). Discrete element modeling of aggregate shape and internal structure effects on Weibull distribution of tensile strength. *Soil and Tillage Research*, 219, 105341. <https://doi.org/10.1016/j.still.2022.105341>
- Barbosa, L. A. P., Keller, T., Ferraz, A. C., & de Oliveira Ferraz, A. C. (2020). Scale effect of aggregate rupture: Using the relationship between friability and fractal dimension to parameterise discrete element models. *Powder Technology*, 375, 327–336. <https://doi.org/10.1016/j.powtec.2020.08.006>
- Bishop, A. (1960). *The principles of effective stress*. Oslo.
- Boivin, P., Garnier, P., & Vauclin, M. (2006). Modeling the soil shrinkage and water retention curves with the same equations. *Soil Science Society of America Journal*, 70, 1082–1093. <https://doi.org/10.2136/sssaj2005.0218>

- Bottinelli, N., Zhou, H., Boivin, P., Zhang, Z. B., Jouquet, P., Hartmann, C., & Peng, X. (2016). Macropores generated during shrinkage in two paddy soils using X-ray micro-computed tomography. *Geoderma*, 265, 78–86. <https://doi.org/10.1016/j.geoderma.2015.11.011>
- Boyle, M., Frankenberger, W. T., Jr., & Stolzy, L. H. (1989). The influence of organic matter on soil aggregation and water infiltration. *Journal of Production Agriculture*, 2, 290–299. <https://doi.org/10.2134/jpa1989.0290>
- Braudeau, E. (1988). Équation généralisée des courbes de retrait d'échantillons de sol structures. *Comptes rendus de l'Académie des Sciences*, 307, 1731–1734.
- Braudeau, E., Costantini, J. M., Bellier, G., & Colleuille, H. (1999). New device and method for soil shrinkage curve measurement and characterization. *Soil Science Society of America Journal*, 63, 525–535. <https://doi.org/10.2136/sssaj1999.03615995006300030015x>
- Braudeau, E., Frangi, J.-P., & Mohtar, R. H. (2004). Characterizing non-rigid aggregated soil-water medium using its shrinkage curve. *Soil Science Society of America Journal*, 68, 359–370. <https://doi.org/10.2136/sssaj2004.3590>
- Capowiez, Y., Gilbert, F., Vallat, A., Poggiale, J. C., & Bonzom, J. M. (2021). Depth distribution of soil organic matter and burrowing activity of earthworms—Mesocosm study using X-ray tomography and luminophores. *Biology and Fertility of Soils*, 57, 337–346. <https://doi.org/10.1007/s00374-020-01536-y>
- Caulk, R. A. (2020). Modeling acoustic emissions in heterogeneous rocks during element method. *Open Geomechanics*, 2, 2. <https://doi.org/10.5802/ogeo.5>
- Chareyre, B., Cortis, A., Catalano, E., & Barthélemy, E. (2012). Pore-scale modeling of viscous flow and induced forces in dense sphere packings. *Transport in Porous Media*, 94, 595–615. <https://doi.org/10.1007/s11242-012-0057-2>
- Cil, M. B., & Alshibli, K. A. (2012). 3D assessment of fracture of sand particles using discrete element method. *Géotechnique Letters*, 2, 161–166. <https://doi.org/10.1680/geolett.12.00024>
- Coppola, A., Comegna, A., Dragonetti, G., Gerke, H. H., & Basile, A. (2015). Simulated preferential water flow and solute transport in shrinking soils. *Vadose Zone Journal*, 14, 1–22. <https://doi.org/10.2136/vzj2015.02.0021>
- Coppola, A., Gerke, H. H., Comegna, A., Basile, A., & Comegna, V. (2012). Dual-permeability model for flow in shrinking soil with dominant horizontal deformation. *Water Resources Research*, 48, W08527. <https://doi.org/10.1029/2011WR011376>
- Costanza-Robinson, M. S., Estabrook, B. D., & Fouhey, D. F. (2011). Representative elementary volume estimation for porosity, moisture saturation, and air-water interfacial areas in unsaturated porous media: Data quality implications. *Water Resources Research*, 47, W07513. <https://doi.org/10.1029/2010WR009655>
- Czarnes, S., Hallett, P. D., Bengough, A. G., & Young, I. M. (2000). Root- and microbial-derived mucilages affect soil structure and water transport. *European Journal of Soil Science*, 51, 435–443. <https://doi.org/10.1046/j.1365-2389.2000.00327.x>
- Durner, W. (1994). Hydraulic conductivity estimation for soils with heterogeneous pore structure. *Water Resources Research*, 30, 211–223. <https://doi.org/10.1029/93WR02676>
- Ellerbrock, R. H., & Gerke, H. H. (2013). Characterization of organic matter composition of soil and flow path surfaces based on physicochemical principles—A review. In D. L. Sparks (Ed.), *Advances in agronomy* (1st ed., Vol. 121, pp. 117–177). Elsevier Inc. <https://doi.org/10.1016/B978-0-12-407685-3.00003-7>
- Fér, M., & Kodešová, R. (2012). Estimating hydraulic conductivities of the soil aggregates and their clay-organic coatings using numerical inversion of capillary rise data. *Journal of Hydrology*, 468–469, 229–240. <https://doi.org/10.1016/j.jhydrol.2012.08.037>
- Garnier, P., Rieu, M., Boivin, P., Vauclin, M., & Baveye, P. (1997). Determining the hydraulic properties of a swelling soil from a transient evaporation experiment. *Soil Science Society of America Journal*, 61, 1555–1563. <https://doi.org/10.2136/sssaj1997.03615995006100060003x>
- Gerke, H. H., & Köhne, J. M. (2002). Estimating hydraulic properties of soil aggregate skins from sorptivity and water retention. *Soil Science Society of America Journal*, 66, 26–36. <https://doi.org/10.2136/sssaj2002.2600>
- Gerke, H. H., Vogel, H., Weber, T. K. D., van der Meij, W. M., & Scholten, T. (2022). 3–4D soil model as challenge for future soil research: Quantitative soil modeling based on the solid phase. *Journal of Plant Nutrition and Soil Science*, 185, 720–744. <https://doi.org/10.1002/jpln.202200239>
- Grayson, R., & Blöschl, G. (2000). *Spatial patterns in catchment hydrology*. Cambridge University Press.
- Haas, C., Leue, M., Ellerbrock, R. H., & Gerke, H. H. (2020). Macropore–matrix mass transfer of reactive solutes quantified by fluorescence imaging. *Vadose Zone Journal*, 19, e20078. <https://doi.org/10.1002/vzj2.20078>
- Harter, T., & Hopmans, J. W. (2004). Role of vadose-zone flow processes in regional-scale hydrology: Review, opportunities, and challenges. In R. A. Feddes, G. H. de Rooij, & J. C. van Dam (Eds.), *International symposium on unsaturated zone modelling: Progress, challenges and applications* (pp. 179–208). Kluwer Academic Publishers.
- Horn, R. (2004). Time dependence of soil mechanical properties and pore functions for arable soils. *Soil Science Society of America Journal*, 68, 1131–1137. <https://doi.org/10.2136/sssaj2004.1131>
- Jarvis, N., Koestel, J., & Larsbo, M. (2016). Understanding preferential flow in the vadose zone: Recent advances and future prospects. *Vadose Zone Journal*, 15, 1–11. <https://doi.org/10.2136/vzj2016.09.0075>
- Legland, D., Arganda-Carreras, I., & Andrey, P. (2016). MorphoLibJ: Integrated library and plugins for mathematical morphology with ImageJ. *Bioinformatics*, 32, 3532–3534. <https://doi.org/10.1093/bioinformatics/btw413>
- Leue, M., Holz, M., Gerke, H. H., Taube, R., Puppe, D., & Wirth, S. (2021). Spatially-distributed microbial enzyme activities at intact, coated macropore surfaces in Luvisol Bt-horizons. *Soil Biology & Biochemistry*, 156, 108193. <https://doi.org/10.1016/j.soilbio.2021.108193>
- Mualem, Y. (1976). A new model for predicting the hydraulic conductivity of unsaturated porous media. *Water Resources Research*, 12, 513–522.
- Nimmo, J. R. (2021). The processes of preferential flow in the unsaturated zone. *Soil Science Society of America Journal*, 85, 1–27. <https://doi.org/10.1002/saj2.20143>
- Peng, X., Hallett, P. D., Zhang, B., & Horn, R. (2011). Physical response of rigid and non-rigid soils to analogues of biological exudates.

- European Journal of Soil Science*, 62, 676–684. <https://doi.org/10.1111/j.1365-2389.2011.01383.x>
- Peng, X., & Horn, R. (2005). Modeling soil shrinkage curve across a wide range of soil types. *Soil Science Society of America Journal*, 69, 584–592. <https://doi.org/10.2136/sssaj2004.0146>
- Peng, X., & Horn, R. (2013). Identifying six types of soil shrinkage curves from a large set of experimental data. *Soil Science Society of America Journal*, 77, 372–381. <https://doi.org/10.2136/sssaj2011.0422>
- Pion, S., & Teillaud, M. (2011). 3D triangulations. In *CGAL user and reference manual*. CGAL Editorial Board.
- Priesack, E., & Durner, W. (2006). Closed-form expression for the multimodal unsaturated conductivity function. *Vadose Zone Journal*, 5, 121–124. <https://doi.org/10.2136/vzj2005.0066>
- Qian, J., Lin, Z., & Shi, Z. (2022). Experimental and modeling study of water-retention behavior of fine-grained soils with dual-porosity structures. *Acta Geotechnica*, 17, 3245–3258. <https://doi.org/10.1007/s11440-022-01483-y>
- Rainer, H., Xinhua, P., Heiner, F., & Jose, D. (2014). Pore rigidity in structured soils—Only a theoretical boundary condition for hydraulic properties? *Soil Science and Plant Nutrition*, 60, 3–14. <https://doi.org/10.1080/00380768.2014.886159>
- Rieckh, H., Gerke, H. H., & Sommer, M. (2012). Hydraulic properties of characteristic horizons depending on relief position and structure in a hummocky glacial soil landscape. *Soil and Tillage Research*, 125, 123–131. <https://doi.org/10.1016/j.still.2012.07.004>
- Schäffer, B., Schulin, R., & Boivin, P. (2013). Shrinkage properties of repacked soil at different states of uniaxial compression. *Soil Science Society of America Journal*, 77, 1930–1943. <https://doi.org/10.2136/sssaj2013.01.0035>
- Schäffer, B., Stauber, M., Mueller, T. L., Müller, R., & Schulin, R. (2008). Soil and macro-pores under uniaxial compression. I. Mechanical stability of repacked soil and deformation of different types of macro-pores. *Geoderma*, 146, 183–191. <https://doi.org/10.1016/j.geoderma.2008.05.019>
- Schnepf, A., Carminati, A., Ahmed, M. A., Ani, M., Benard, P., Bentz, J., Bonkowski, M., Knott, M., Diehl, D., Duddek, P., Kröner, E., Javaux, M., Landl, M., Lehnendorff, E., Lippold, E., Lieu, A., Mueller, C. W., Oburger, E., Otten, W., ... Vetterlein, D. (2022). Linking rhizosphere processes across scales: Opinion. *Plant and Soil*, 478, 5–42. <https://doi.org/10.1007/s11104-022-05306-7>
- Scholtès, L., & Donzé, F.-V. (2013). A DEM model for soft and hard rocks: Role of grain interlocking on strength. *Journal of the Mechanics and Physics of Solids*, 61, 352–369. <https://doi.org/10.1016/j.jmps.2012.10.005>
- Scholtès, L., & Donzé, F. V. (2015). A DEM analysis of step-path failure in jointed rock slopes. *Comptes Rendus Mécanique*, 343, 155–165. <https://doi.org/10.1016/j.crme.2014.11.002>
- Seki, K., Toride, N., & van Genuchten, M. Th. (2022). Closed-form hydraulic conductivity equations for multimodal unsaturated soil hydraulic properties. *Vadose Zone Journal*, 21, e20168. <https://doi.org/10.1002/vzj2.20168>
- Šmilauer, V., Chareyre, B., Duriez, J., Eulitz, A., Gladky, A., Guo, N., Jakob, C., Kozicki, J., Kneib, F., Modenese, C., Stranský, J., & Thoeni, K. (2015). Using and programming. In *Yade documentation* (2nd ed.). Zenodo. <https://doi.org/10.5281/zenodo.34043>
- Stange, C. F., & Horn, R. (2005). Modeling the soil water retention curve for conditions of variable porosity. *Vadose Zone Journal*, 4, 602–613. <https://doi.org/10.2136/vzj2004.0150>
- Tang, C. S., Zhu, C., Cheng, Q., Zeng, H., Xu, J. J., Tian, B. G., & Shi, B. (2021). Desiccation cracking of soils: A review of investigation approaches, underlying mechanisms, and influencing factors. *Earth-Science Reviews*, 216, 103586. <https://doi.org/10.1016/j.earscirev.2021.103586>
- Tisdall, J. M., & Oades, J. M. (1982). Organic matter and water-stable aggregates in soils. *Journal of Soil Science*, 33, 141–163. <https://doi.org/10.1111/j.1365-2389.1982.tb01755.x>
- van der Linden, J. H., Tordesillas, A., & Narsilio, G. A. (2019). Preferential flow pathways in a deforming granular material: Self-organization into functional groups for optimized global transport. *Scientific Reports*, 9, 18231. <https://doi.org/10.1038/s41598-019-54699-6>
- van Genuchten, M. Th. (1980). A closed-form equation for predicting the hydraulic conductivity of unsaturated soils. *Soil Science Society of America Journal*, 44, 892–898. <https://doi.org/10.2136/sssaj1980.03615995004400050002x>
- van Genuchten, M. Th., & Nielsen, R. (1985). On describing and predicting the hydraulic properties of unsaturated soils. *Annales Geophysicae*, 3, 615–628.
- Vereecken, H., Kasteel, R., Vanderborght, J., & Harter, T. (2007). Upscaling hydraulic properties and soil water flow processes in heterogeneous soils: A review. *Vadose Zone Journal*, 6, 1–28. <https://doi.org/10.2136/vzj2006.0055>
- Vereecken, H., Schnepf, A., Hopmans, J. W., Javaux, M., Or, D., Roose, T., Vanderborght, J., Young, M. H., Amelung, W., Aitkenhead, M., Allison, S. D., Assouline, S., Baveye, P., Berli, M., Brüggemann, N., Finke, P., Flury, M., Gaiser, T., Govers, G., ... Young, I. M. (2016). Modeling soil processes: Review, key challenges, and new perspectives. *Vadose Zone Journal*, 15, vzj2015.09.0131. <https://doi.org/10.2136/vzj2015.09.0131>
- Vetterlein, D., Carminati, A., Kögel-Knabner, I., Bienert, G. P., Smalla, K., Oburger, E., Schnepf, A., Banitz, T., Tarkka, M. T., & Schlüter, S. (2020). Rhizosphere spatiotemporal organization—A key to rhizosphere functions. *Frontiers in Agronomy*, 2, 8. <https://doi.org/10.3389/fagro.2020.00008>
- Virtanen, P., Gommers, R., Oliphant, T. E., Haberland, M., Reddy, T., Cournapeau, D., Burovski, E., Peterson, P., Weckesser, W., Bright, J., van der Walt, S. J., Brett, M., Wilson, J., Millman, K. J., Mayorov, N., Nelson, A. R. J., Jones, E., Kern, R., Larson, E., ... Vázquez-Baeza, Y. (2020). SciPy 1.0: Fundamental algorithms for scientific computing in Python. *Nature Methods*, 17, 261–272. <https://doi.org/10.1038/s41592-019-0686-2>
- Vogel, H. J., Gerke, H. H., Mitrach, R., Zahl, R., & Wöhling, T. (2023). Soil hydraulic conductivity in the state of nonequilibrium. *Vadose Zone Journal*, 22, e20238. <https://doi.org/10.1002/vzj2.20238>
- Vogel, T., Brezina, J., Dohnal, M., & Dusek, J. (2010). Physical and numerical coupling in dual-continuum modeling of preferential flow. *Vadose Zone Journal*, 9, 260–267. <https://doi.org/10.2136/vzj2009.0091>
- Yuan, C., & Chareyre, B. (2017). A pore-scale method for hydromechanical coupling in deformable granular media. *Computer Methods in Applied Mechanics and Engineering*, 318, 1066–1079. <https://doi.org/10.1016/j.cma.2017.02.024>
- Yuan, C., & Chareyre, B. (2022). Transitioning from the funicular to the pendular regime in granular soils. *Géotechnique*, 72, 825–831. <https://doi.org/10.1680/jgeot.20.P.141>
- Yuan, C., Chareyre, B., & Darve, F. (2018). Deformation and stresses upon drainage of an idealized granular material. *Acta Geotechnica*, 13, 961–972. <https://doi.org/10.1007/s11440-017-0601-x>

Zarebanadkouki, M., Ahmed, M. A., & Carminati, A. (2016). Hydraulic conductivity of the root-soil interface of lupin in sandy soil after drying and rewetting. *Plant and Soil*, 398, 267–280. <https://doi.org/10.1007/s11104-015-268-1>

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

How to cite this article: Barbosa, L. A. P., & Gerke, H. H. (2024). Coupled hydro-mechanical pore-scale modeling of biopore-coated clods for upscaling soil shrinkage and hydraulic properties. *Vadose Zone Journal*, 23, e20325.

<https://doi.org/10.1002/vzj2.20325>